

SQL Interview Notes

RAG-friendly interview preparation material covering core concepts and explanations.

1. SQL and RDBMS

SQL stands for Structured Query Language and is used to work with relational databases. An RDBMS stores data in related tables and commonly uses SQL for database operations.

2. DDL

Data Definition Language commands define or change database structures. CREATE creates objects, ALTER changes their structure, DROP removes an object, and TRUNCATE removes all rows while keeping the table structure.

3. DML

Data Manipulation Language is used to modify table data. INSERT adds rows, UPDATE changes existing rows, and DELETE removes selected rows.

4. SELECT and WHERE

SELECT retrieves data. WHERE filters rows according to a condition before grouping and aggregation are applied.

5. ORDER BY

ORDER BY sorts query results. Sorting can be ascending or descending and can use one or more columns.

6. GROUP BY and HAVING

GROUP BY creates groups of rows for aggregate calculations. HAVING filters groups after aggregation, while WHERE filters individual rows before grouping.

7. Aggregate Functions

COUNT counts rows or values, SUM calculates a total, AVG calculates an average, MIN finds the smallest value, and MAX finds the largest value.

8. Operators

AND requires multiple conditions to be true, OR allows either condition, and NOT reverses a condition. LIKE supports pattern matching, BETWEEN checks a range, and IN checks membership in a list.

9. Keys and Constraints

PRIMARY KEY uniquely identifies rows. FOREIGN KEY represents a relationship with another table. UNIQUE prevents duplicate values. NOT NULL requires a value. CHECK enforces a condition. DEFAULT supplies a value when one is not provided.

10. Joins

INNER JOIN returns matching rows. LEFT JOIN keeps all rows from the left table. RIGHT JOIN keeps all rows from the right table. A join condition commonly uses related key columns.

11. Subqueries

A subquery is a query nested inside another query. It can be used in SELECT, FROM, WHERE, or HAVING depending on the requirement.

12. Views and Indexes

A view stores a query definition and presents its result like a table. An index can improve lookup speed on suitable columns but adds storage and modification overhead.

13. Transactions

A transaction groups related database operations. COMMIT permanently saves a transaction. ROLLBACK undoes changes that have not been committed. SAVEPOINT creates a point to which a transaction can roll back.

14. DROP, DELETE and TRUNCATE

DROP removes the database object itself. DELETE removes selected rows and can use WHERE. TRUNCATE removes all rows while preserving the table structure. Exact transactional behavior can vary by database system.

15. SQL Query Practice Concepts

Interview questions commonly combine SELECT, WHERE, JOIN, GROUP BY, HAVING, ORDER BY, aggregate functions, subqueries, and constraints. Understanding the order and purpose of these clauses is important.